Quantifying Attention Flow in Transformers

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Transformer Models

Basics

Proposed by famous Google paper *Attention Is All You Need* [2].

- Numerous applications in NLP, Computer Vision, ...
- Process sequence of input **tokens**
- Stack of identical layers
- Each applying **Self-Attention**

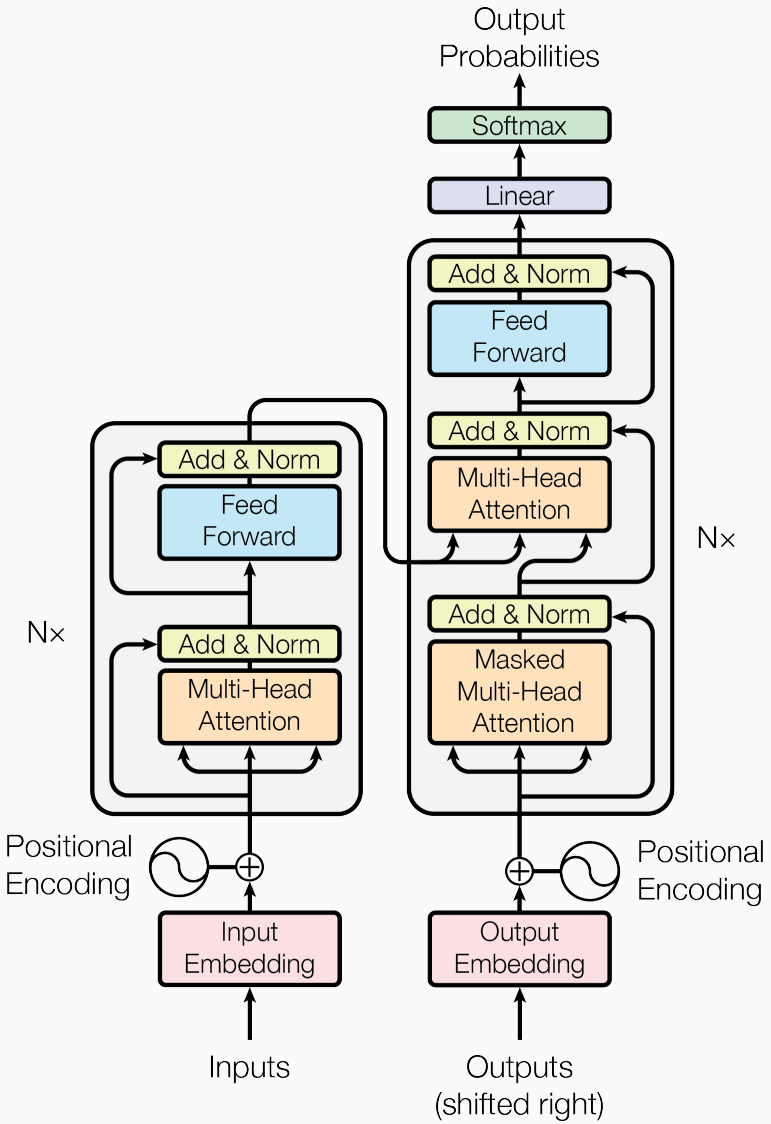

Figure 1: Transformer Model. Taken from [2].

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Result

After this operation, each token captures **contextual information from all other tokens**.

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How can we quantify this information flow across multiple layers?

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Verb Number Prediction Task

Predicting singularity or plurality of a verb of a sentence, when the input is the sentence up to the verb position

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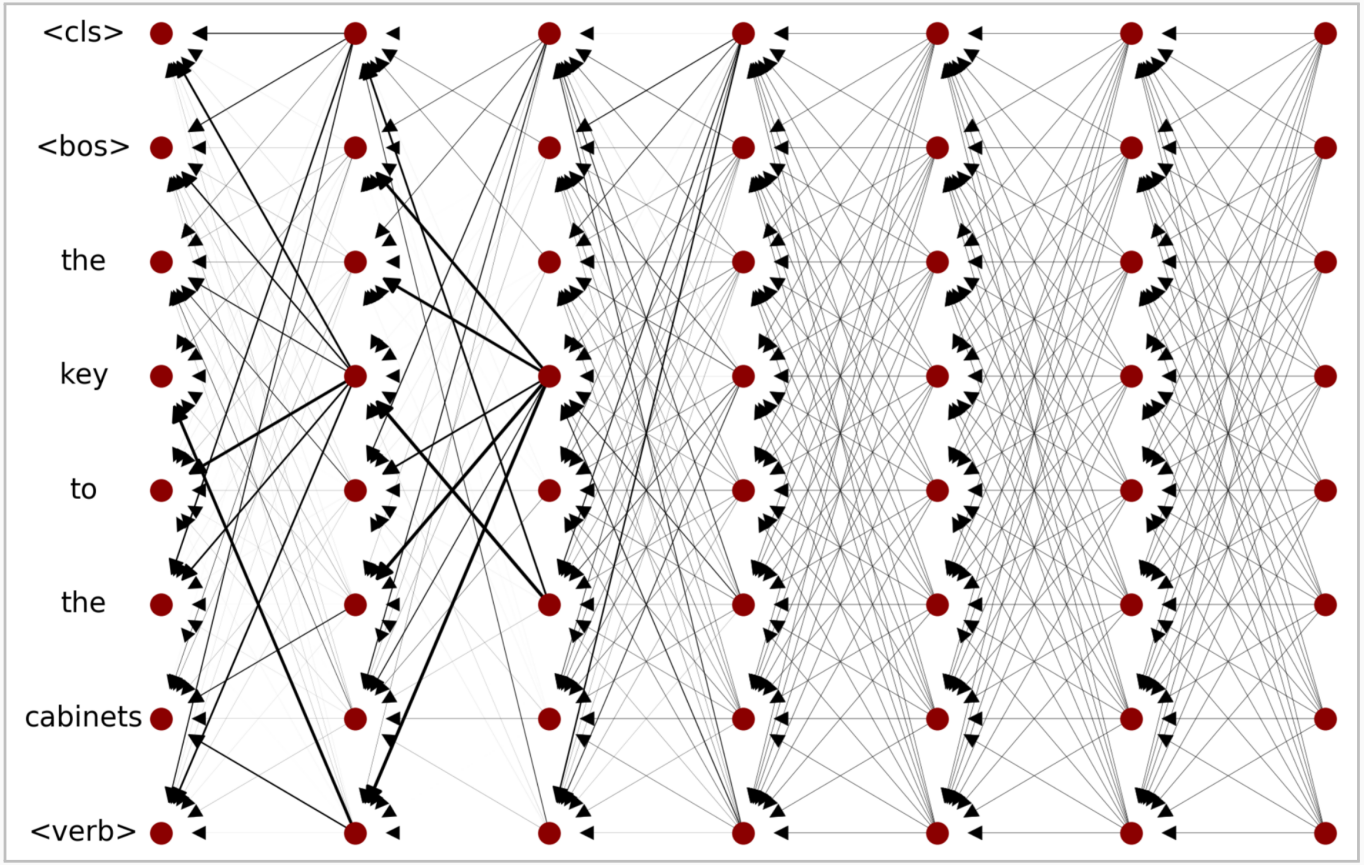

Figure 2: Attention Flow Graph of input sentence.

Raw Attention Visualized

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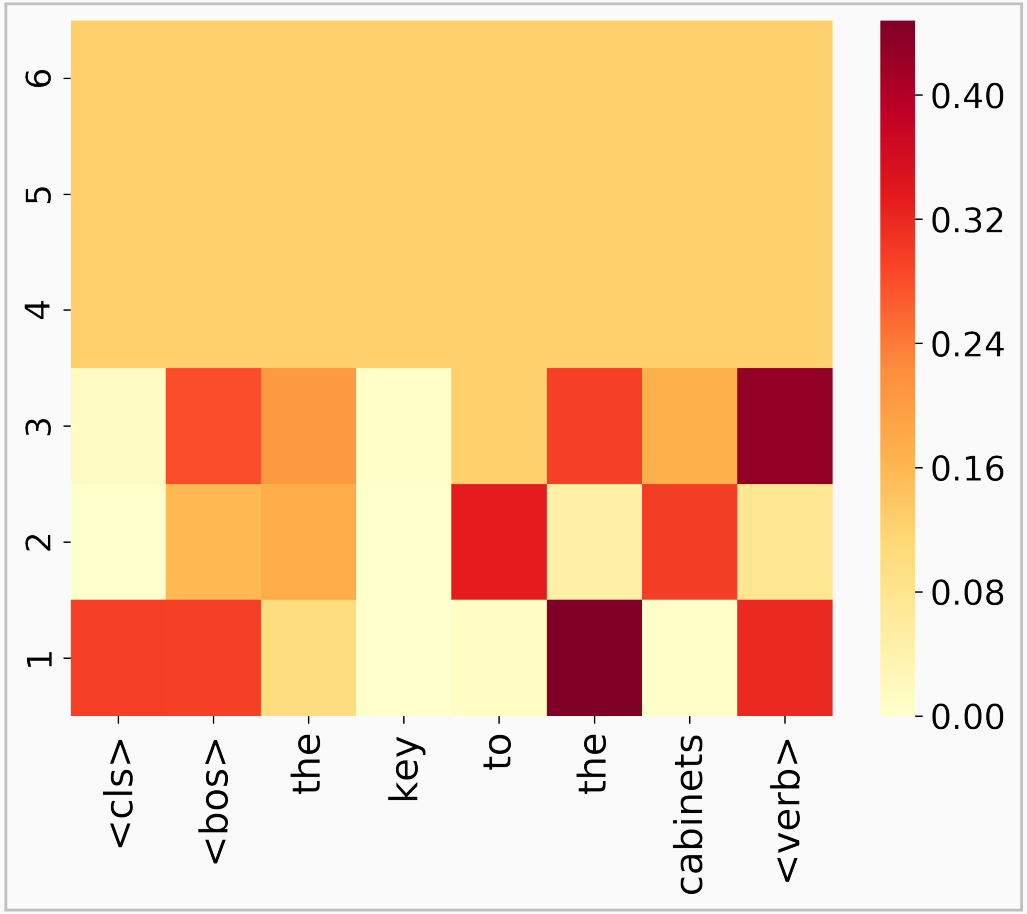

Figure 3: Raw Attention map for the <cls> token at different layers.

Attention Rollout

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- One computation in $\mathcal{O}(d^2 \times n^4)$

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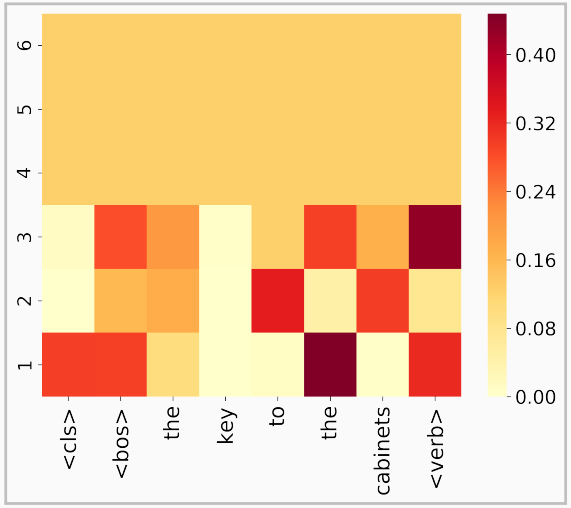

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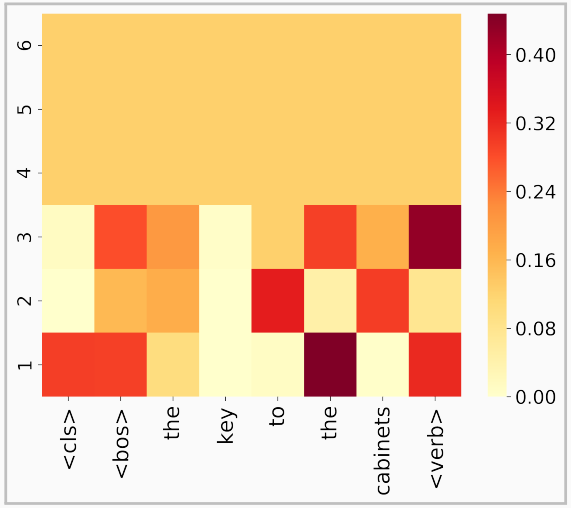

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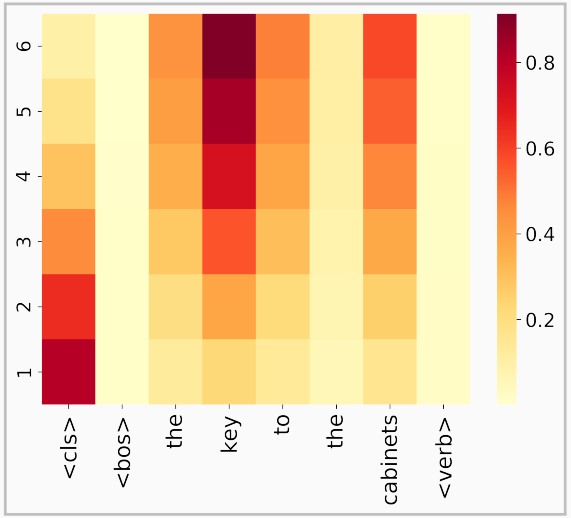

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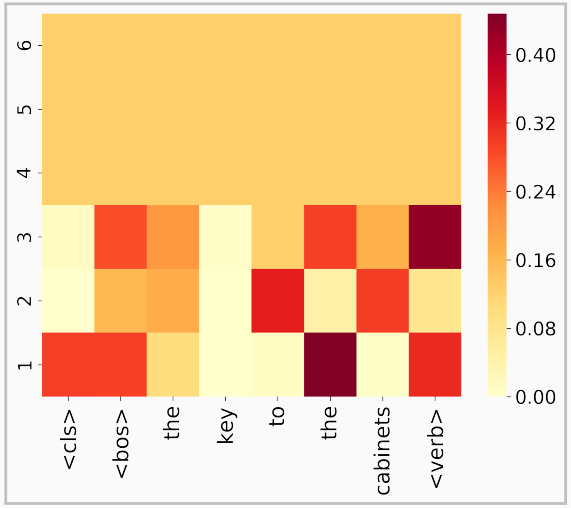

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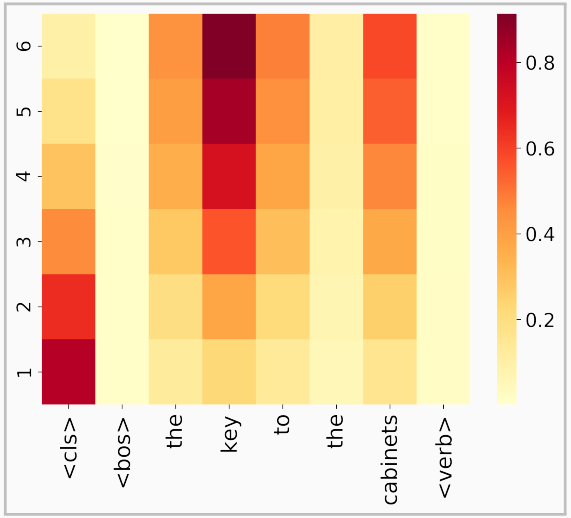

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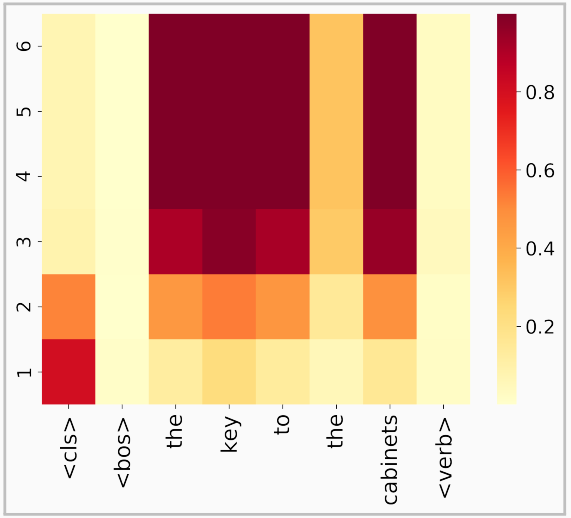

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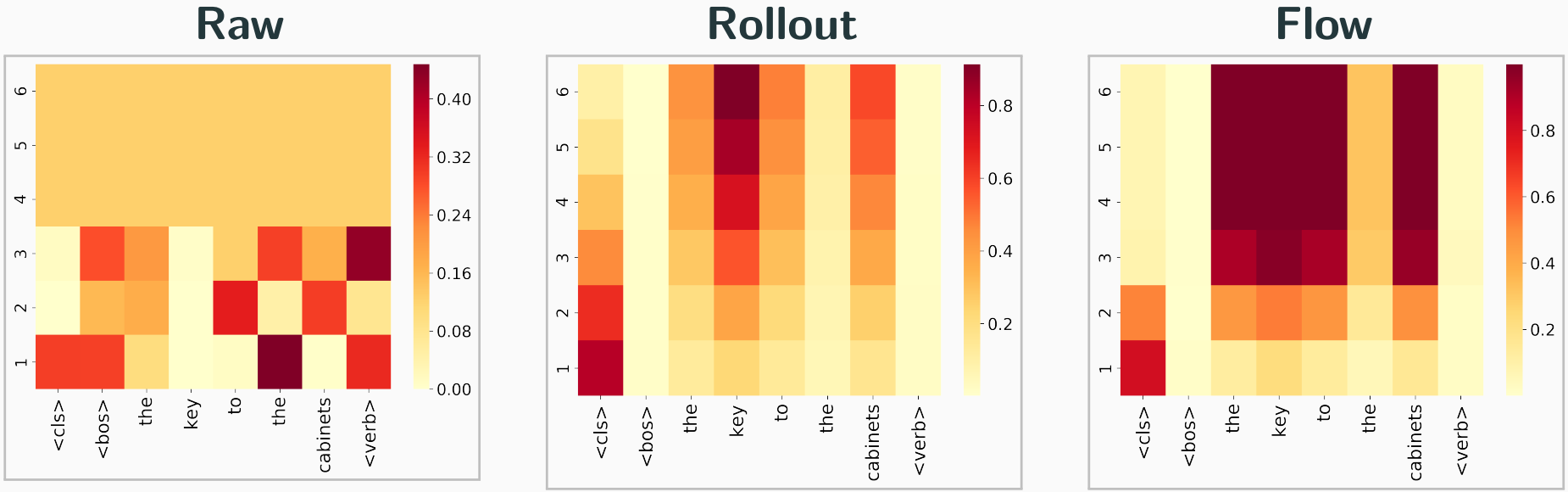

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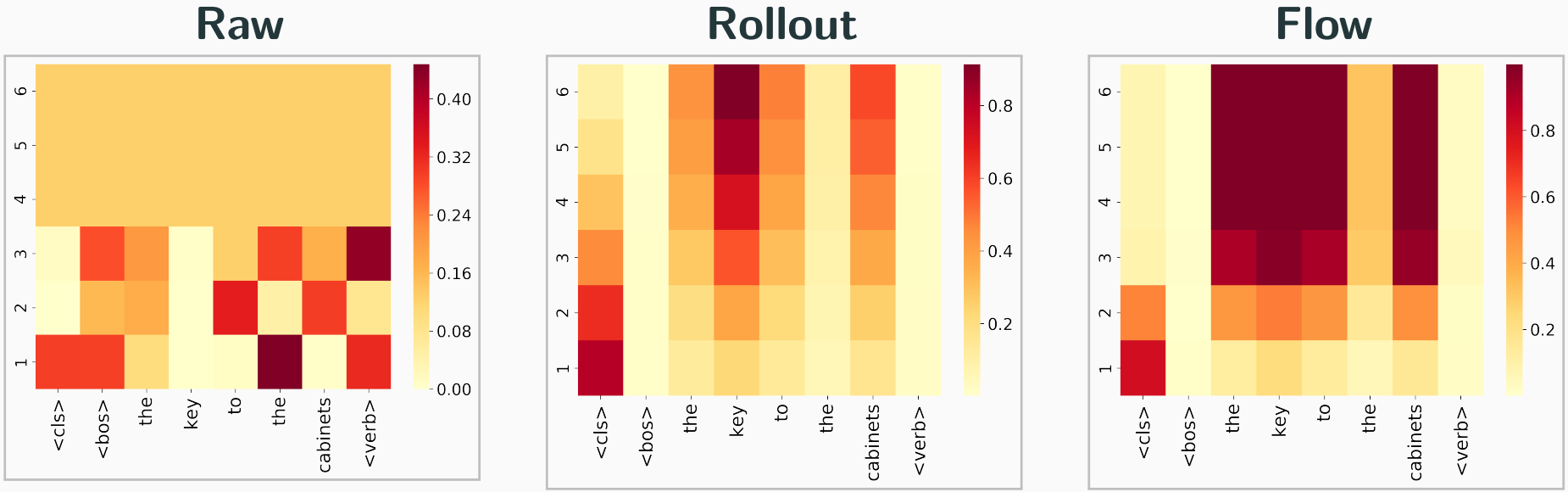

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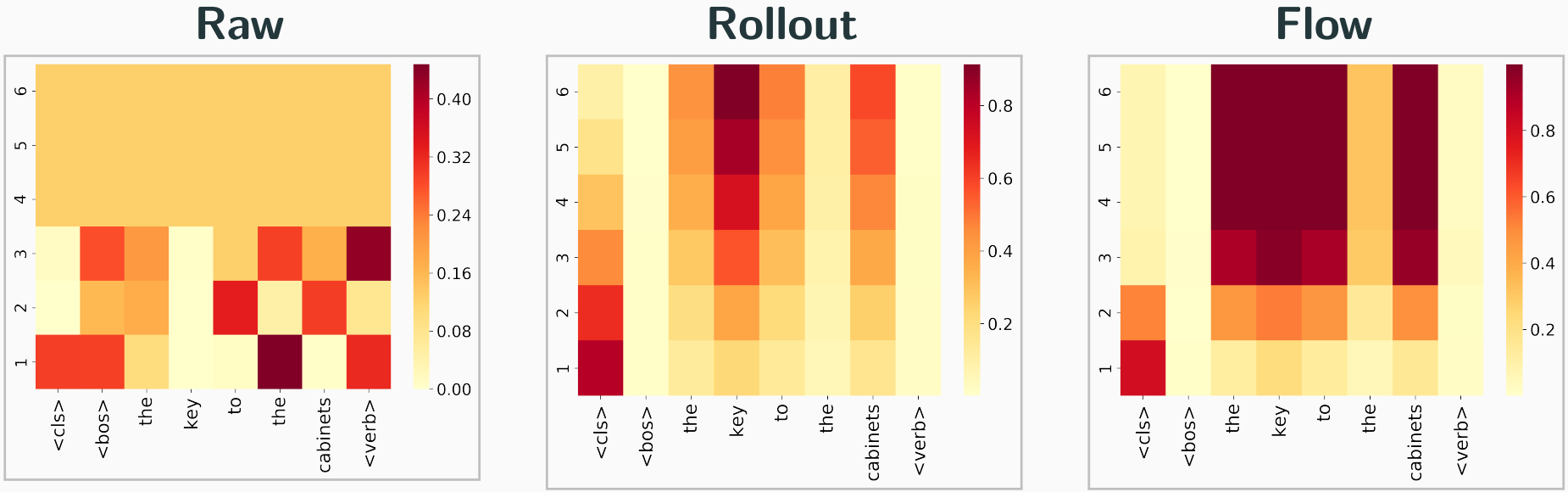

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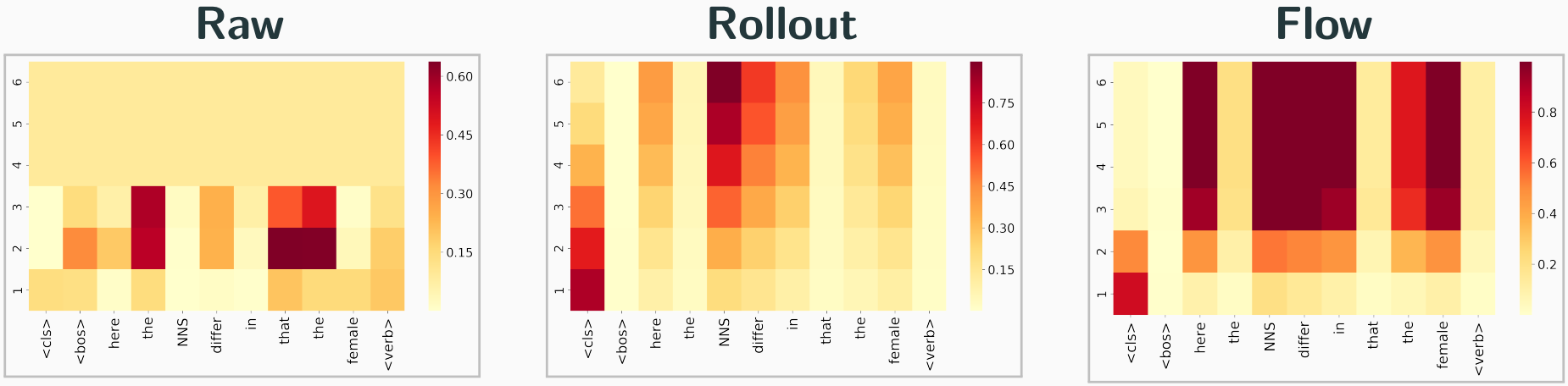

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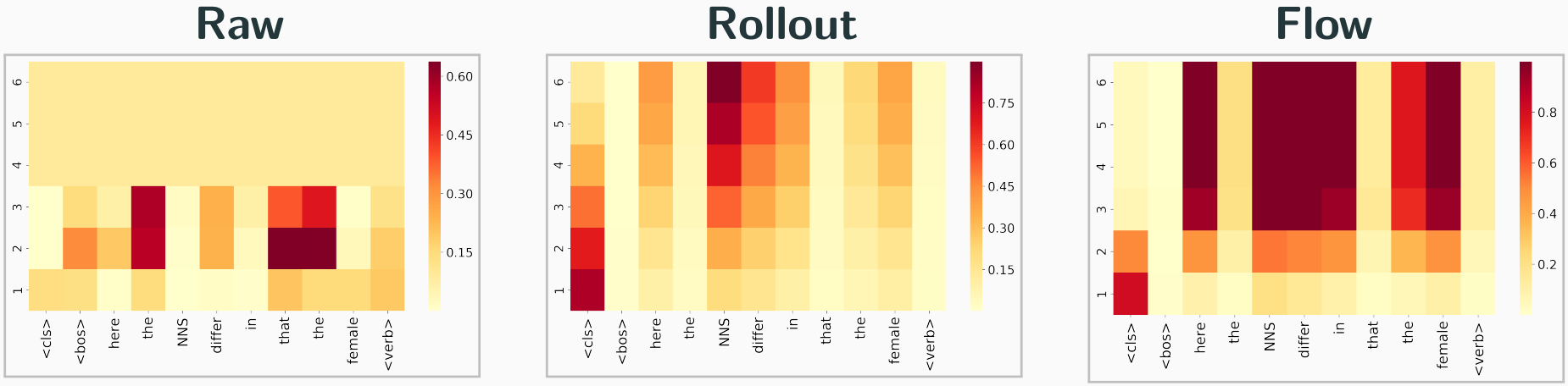

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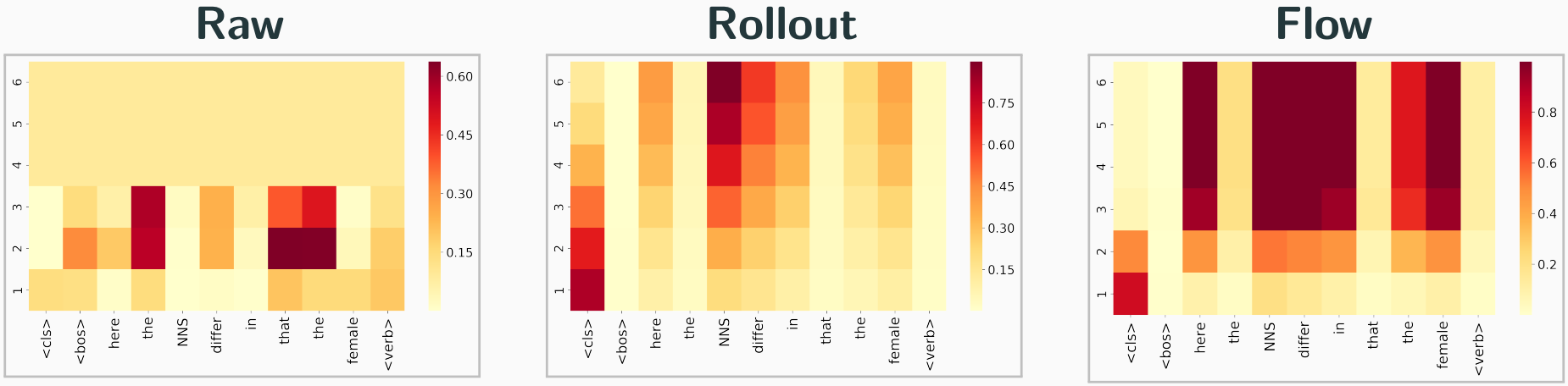

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- Architecture agnostic
- Provide insights into correct classifications and misclassifications
- Attention Rollout computationally more feasible and more accurate results

Thank you for your attention!

Any questions?

References

[1] S. Abnar and W. Zuidema, **"Quantifying attention flow in transformers,"** in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, D. Jurafsky et al., Eds., Online: Association for Computational Linguistics, Jul. 2020, pp. 4190–4197. DOI: 10.18653/v1/2020.acl-main.385. [Online]. Available: https://aclanthology.org/2020.acl-main.385/.

[2] A. Vaswani et al., **Attention is all you need,** 2023. arXiv: 1706.03762 [cs.CL]. [Online]. Available: https://arxiv.org/abs/1706.03762.

[3] A. Radford et al., **"Language models are unsupervised multitask learners,"**, 2019. [Online]. Available: https://api.semanticscholar.org/CorpusID:160025533.

[4] T. Linzen, E. Dupoux, and Y. Goldberg, **Assessing the ability of lstms to learn syntax-sensitive dependencies,**

Setup

Model

- Transformer encoder model with 6 layers
- GPT-2 blocks from [3]
- Embedding of `<cls>` as input to classifier
- 96% final accuracy on verb number prediction task

Dataset

- Public subject-verb agreement dataset [4]